

POSITION STATEMENT

Laser treatment of hyperpigmented lesions: position statement of the European Society of Laser in Dermatology

Abstract

Background

Lasers and intense pulsed light sources (IPLS) are widely used for the treatment of various pigmentary disorders. They are sometimes perceived as universal solutions capable of eliminating any pigmented lesion. While they represent the treatment of choice for several hyperpigmented conditions, they can exacerbate certain disorders and carry potential adverse effects.

Objective

To provide evidence-based recommendations on the use of lasers and IPLS in the management of hyperpigmented lesions.

Methods

These guidelines were developed by a consensus panel of experts in pigment laser surgery on behalf of the European Society for Laser Dermatology. Recommendations were formulated based on the quality of available evidence regarding efficacy, safety, tolerability, cosmetic outcomes, and patient satisfaction or preference. In cases of limited evidence, expert opinion was considered.

Results

Lasers and IPLS demonstrate high efficacy in treating numerous hyperpigmented lesions, including lentigos, dermal hypermelanocytosis, and heavy metal deposits. However, caution is required when treating conditions such as café au lait macules, melasma, and postinflammatory hyperpigmentation, where risks of recurrence or worsening are significant. Accurate diagnosis is essential. When laser or IPL therapy is indicated, optimal wavelength and treatment parameters must be selected according to skin phototype, and the depth and origin of the pigment.

Conclusion

Despite their potential effectiveness, lasers and IPLS are not suitable for all types of hyperpigmented lesions. A precise diagnosis is mandatory to determine whether laser-based approaches are appropriate or if alternative treatments should be pursued.

Introduction

Skin color arises from pigments located in the epidermis and dermis. Melanins—primarily eumelanin (dark brown, predominant in darker skin types) and pheomelanin (red to light brown, more common in fair skin)—are the principal pigments in human skin. Other endogenous pigments, such as haemoglobin and bilirubin, also contribute to skin coloration. Pigmentary disorders are among the most common dermatological conditions, affecting up to 60% of individuals in some populations. These disorders may be inherited or acquired and manifest as hyperpigmentation, hypopigmentation, or abnormal skin coloration.

Most pigmentary disorders result from quantitative or qualitative abnormalities in melanin production or distribution. However, alterations in other endogenous pigments or deposition of exogenous substances can also lead to dyschromia. Effective management, particularly laser treatment, requires thorough understanding of the underlying pathophysiology, the anatomical location of the pigment (epidermal vs. dermal), and the specific chromophore involved (Fig. 1).

Methods

A comprehensive literature review was conducted covering publications from 1983 to April 2018 (see Appendix S1 for detailed search strategy). The initial search yielded 8,661 results. After screening, 981 relevant articles—including case reports, clinical trials, observational studies, randomized controlled trials, and systematic reviews—were selected and critically appraised. Articles available only as abstracts were excluded. Additional references were identified through cross-checking bibliographies of selected papers.

All selected studies were analyzed by the Consensus Panel (TP, RG, LS, TF, GK, HJL, LM, AB). The evaluation focused on answering key clinical questions for each hyperpigmentary disorder:

- (i) Which lasers or light sources are indicated?
- (ii) What level of therapeutic efficacy is achievable, and how was it measured?
- (iii) Are there comparative studies between different lasers or light sources?
- (iv) What are the associated side effects?

A draft of the recommendations was circulated among panel members for review, feedback, and revision. Multiple rounds of discussion were conducted until full consensus was achieved.